

D338

Cloud Platform Solutions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A healthcare startup with 35 analysts, most of whom joined this year, needs every analyst to read data in one storage account. The cloud admin, who is also studying for a certification, is deciding how to grant access. What is the best approach?

- A. Assign the read role to each of the 35 analysts one by one.
- B. Give access to new hires only after they pass a certification.
- C. Put the analysts in a group and give the group a read role on that storage account only.
- D. Give each analyst the owner role on the whole subscription.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A law firm connects its email, document system, and billing app to one identity provider. Staff sign in once each morning and reach all three. The firm has offices in two time zones. When a staff member opens the billing app, what verifies who she is?

- A. The document system, since she used it first.
- B. The identity provider, through a token the billing app trusts.
- C. The billing app, checking its own stored password.
- D. A separate sign-in for each time zone.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An online store's web tier runs at 99.9% availability and depends on a database service at 99.95%. Both run in a region that recently opened a second data center. If either one is down, the store is down. What is the store's expected availability?

- A. 99.95%
- B. 99.9%
- C. About 99.99995%
- D. About 99.85%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Two identical web servers, each 99% available, are placed behind a load balancer so that either one alone can serve all traffic. Ignoring the load balancer, what is the combined availability?

- A.** 98.01%
- B.** 99%
- C.** 100%, because there is a spare.
- D.** 99.99%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A ticket-sales site gets huge traffic spikes when concerts are announced. The operations lead, a former musician, wants to handle spikes and survive the loss of any single server. Which approach fits best?

- A.** Scale out with an autoscaling group of several smaller instances.
- B.** Scaling is not needed if the VM is always large enough.
- C.** Scale up to the largest single VM size.
- D.** Add more disk to the current VM.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

An admin changes a VM from 4 vCPUs to 16 vCPUs. What kind of scaling is this?

- A.** Load balancing.
- B.** Autoscaling.
- C.** Horizontal scaling (scale out).
- D.** Vertical scaling (scale up).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A research lab stores 80 TB of instrument data that must be kept for ten years for grant compliance and is expected to be read only if an audit occurs. The lab's director recently published a major paper. Which storage tier fits best?

- A. Hot tier.
- B. Local VM disks.
- C. Archive tier.
- D. Cool tier, because it is always the cheapest option.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A company moved its product-image library to the archive tier to save money. The images are loaded every time a customer views the website. What problem results?

- A. Images must be rehydrated before reading, causing long delays and retrieval charges.
- B. Storage costs go up.
- C. No problem; all tiers read at the same speed.
- D. The images are deleted after 30 days.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A marketing agency must upload campaign files into a client's storage container over the next week. The agency has worked with the client for six years. How should the client grant access?

- A. Send the storage account key.
- B. Share the account key because the relationship is long-standing.
- C. Issue a write-only access token scoped to that container that expires in one week.
- D. Make the container public.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A company has three virtual networks: Hub, App, and Data. Hub is peered with App, and Hub is peered with Data. There are no other peerings or custom routes. The networks were set up during a busy quarter-end. Can a VM in App reach a VM in Data directly?

- A. Yes, because both are peered with Hub.
- B. No; App and Data need their own peering or routing through a hub device.
- C. Only if a storage account is shared.
- D. Yes, as long as the VMs are in the same region.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Two peered virtual networks are connected, but web servers in one still cannot reach the database port in the other. What is the most likely cause?

- A. The databases need a public IP address.
- B. A network security rule is blocking that port between the subnets.
- C. Peered networks can never carry database traffic.
- D. Peering is not transitive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A retailer's ops team, which has grown from 3 to 12 people, builds each web server by hand. Servers now differ in patches and settings, and outages are usually reported by customers first. What should the team do?

- A. Deploy from a hardened, versioned golden image and set up metric alerts with thresholds.
- B. Add more staff to the ops team.
- C. Check server logs once a month.
- D. Keep building by hand but write better notes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A healthcare startup with 35 analysts, ~~most of whom joined this year~~, needs every analyst to read data in one storage account. The cloud admin, ~~who is also studying for a certification~~, is deciding how to grant access. What is the best approach?

Target word: access design **Decides:** Many users with the same need, one resource.

A	Dropped element	Works, but drops group management and becomes hard to maintain.
B	Camouflage	Certification status is not an access-design principle.
C	Correct	Group assignment at the narrowest scope is least privilege and easy to manage.
D	Adjacent category	Far broader rights and scope than needed.

2. Answer B Concept 2

A law firm connects its email, document system, and billing app to one identity provider. Staff sign in once each morning and reach all three. ~~The firm has offices in two time zones~~. When a staff member opens the billing app, what verifies who she is?

Target word: what does authentication under SSO **Decides:** The identity provider authenticates and issues a token the app trusts.

A	Confusion set	Apps do not vouch for each other; the provider does.
B	Correct	The app relies on the provider's assertion.
C	Inverted operation	SSO removes app-level password checks.
D	Camouflage	Office time zones do not change authentication.

3. Answer D Concept 3

An online store's web tier runs at 99.9% availability and depends on a database service at 99.95%. ~~Both run in a region that recently opened a second data center~~. If either one is down, the store is down. What is the store's expected availability?

Target word: availability of components in series **Decides:** Both must be up; multiply.

A	Absolute language	Assumes the system gets the best component's SLA.
B	Wrong stem number	Takes the lower SLA without combining.
C	Inverted operation	Uses the parallel formula for components that are in series.
D	Correct	$0.999 \times 0.9995 \approx 0.9985$.

4. Answer D Concept 3 · transfer

Two identical web servers, each 99% available, are placed behind a load balancer so that either one alone can serve all traffic. Ignoring the load balancer, what is the combined availability?

Target word: availability in parallel **Decides:** The service fails only if both fail at once.

A	Inverted operation	Multiplies availabilities as if they were in series.
B	Dropped element	Ignores the benefit of the second server.
C	Absolute language	Both can still fail at the same time.
D	Correct	$1 - (0.01 \times 0.01) = 0.9999$.

5. Answer A Concept 4

A ticket-sales site gets huge traffic spikes when concerts are announced. The operations lead, ~~a former musician~~, wants to handle spikes and survive the loss of any single server. Which approach fits best?

Target word: scaling approach **Decides:** Variable load plus survive single-server loss.

A	Correct	More instances absorb spikes and remove a single point of failure.
B	Absolute language	Over-sizing wastes money and still does not survive failure.
C	Inverted operation	One large VM is still a single point of failure and cannot shrink easily.
D	Adjacent category	Storage size does not add compute or redundancy.

6. Answer D Concept 4 • transfer

An admin changes a VM from 4 vCPUs to 16 vCPUs. What kind of scaling is this?

Target word: name the scaling type **Decides:** One machine got bigger.

A	Confusion set	Distributes traffic; it does not change machine size.
B	Adjacent category	A policy that changes capacity, usually by instance count.
C	Inverted operation	That adds instances, not size.
D	Correct	Resizing one machine is vertical.

7. Answer C Concept 5

A research lab stores 80 TB of instrument data that must be kept for ten years for grant compliance and is expected to be read only if an audit occurs. ~~The lab's director recently published a major paper.~~ Which storage tier fits best?

Target word: tier choice **Decides:** Long retention, very rare reads.

A	Inverted operation	Priced for frequent access; wastes money here.
B	Adjacent category	Not built for durable, low-cost long-term retention.
C	Correct	Cheapest for data that is almost never read.
D	Absolute language	Archive is cheaper for data this rarely read.

8. Answer A Concept 5 • transfer

A company moved its product-image library to the archive tier to save money. The images are loaded every time a customer views the website. What problem results?

Target word: effect of the wrong tier **Decides:** Archive data is not immediately readable.

A	Correct	Archive trades cheap storage for slow, costly access.
B	Inverted operation	Storage cost drops; access is the problem.
C	Absolute language	Tiers differ sharply in access time.
D	Confusion set	Tiering does not delete data.

9. Answer C Concept 6

A marketing agency must upload campaign files into a client's storage container over the next week. ~~The agency has worked with the client for six years.~~ How should the client grant access?

Target word: external access method **Decides:** Outside party, one container, limited time.

A	Adjacent category	Gives full control of the entire account with no expiry.
B	Camouflage	Trust history does not justify full-key sharing.
C	Correct	Limited scope and expiry reduce exposure.
D	Inverted operation	Opens it to everyone instead of limiting access.

10. Answer B Concept 7

A company has three virtual networks: Hub, App, and Data. Hub is peered with App, and Hub is peered with Data. There are no other peerings or custom routes. ~~The networks were set up during a busy quarter end.~~ Can a VM in App reach a VM in Data directly?

Target word: transitivity of peering **Decides:** Peering is not transitive.

A	Inverted operation	Assumes peering passes through.
B	Correct	Peering links only the two networks named.
C	Confusion set	Storage sharing does not connect networks.
D	Dropped element	Region alone does not create a connection.

11. Answer B Concept 7 · transfer

Two peered virtual networks are connected, but web servers in one still cannot reach the database port in the other. What is the most likely cause?

Target word: why traffic fails despite peering **Decides:** Peering provides the path; security rules still filter traffic.

A	Inverted operation	Peering uses private addresses; public exposure is the wrong direction.
B	Correct	Filtering rules apply on top of peering.
C	Absolute language	Peering carries any allowed traffic.
D	Confusion set	Only two networks are involved, so transitivity does not apply.

12. Answer A Concept 8

A retailer's ops team, ~~which has grown from 3 to 12 people,~~ builds each web server by hand. Servers now differ in patches and settings, and outages are usually reported by customers first. What should the team do?

Target word: fix drift and late detection **Decides:** Inconsistent builds and no proactive alerting.

A	Correct	Images remove drift; alerts catch failures before customers do.
B	Camouflage	Team size is not the cause.
C	Adjacent category	Too slow; monitoring needs real-time alerts.
D	Dropped element	Still leaves drift and no alerting.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Group-based access and least privilege	Assign roles to groups at the narrowest scope that does the job, then manage people by group membership.	Granting broad owner rights to individual users at the subscription level for convenience.	Many users needing the same access, or a role assigned far above where it is needed.
2	Single sign-on and federation	In SSO the identity provider authenticates the user and issues a token; the application trusts that token instead of storing its own password.	Thinking each app still checks a separate password under SSO.	Users signing in once to reach several apps.
3	Composite availability	Components in series multiply availabilities (lower total); redundant components in parallel give 1 – (product of failure rates) (higher total).	Assuming the system inherits the best component's SLA.	Two or more availability percentages that depend on each other or back each other up.
4	Scale up vs. scale out	Scaling up adds size to one machine; scaling out adds more machines, which suits variable load and improves resilience.	Calling a larger VM size 'horizontal' scaling.	Changing VM size vs. changing instance count.
5	Storage access tiers	Hot tier suits frequent access; cool suits infrequent access; archive is cheapest to store but slow and costly to retrieve.	Putting data that is read daily in archive to save money.	How often and how fast data must be read.
6	Delegated, time-limited storage access	Give outside parties a scoped, expiring access token (such as a shared access signature) rather than the account key.	Sharing the full storage account key with a partner.	An external party needs limited access for a limited time.
7	Virtual network peering and filtering	Peering connects two virtual networks directly and is not transitive; network security rules filter traffic by source, destination, port, and protocol.	Assuming A-B and B-C peering lets A reach C.	Three networks with two peering links.
8	Standard images and monitoring	Build servers from a hardened, versioned golden image and monitor metrics with alert thresholds so drift and failures are caught early.	Hand-building each server and checking health only after users complain.	Inconsistent servers or failures found by users first.

Course scope (public description): Cloud Platform Solutions examines skills in identifying cloud system administration tasks related to user access groups, single sign-on (SSO), and server deployments. Students will gain skills in determining machine access for cloud storage solutions and in explaining the configuration of virtual machines for availability, scalability, performance, and security. Students will also be introduced to implementing virtual networking services and machine image monitoring. The following courses are prerequisites: Network and Security—Foundations, Network and Security—Applications, Networks, and Cloud Applications.